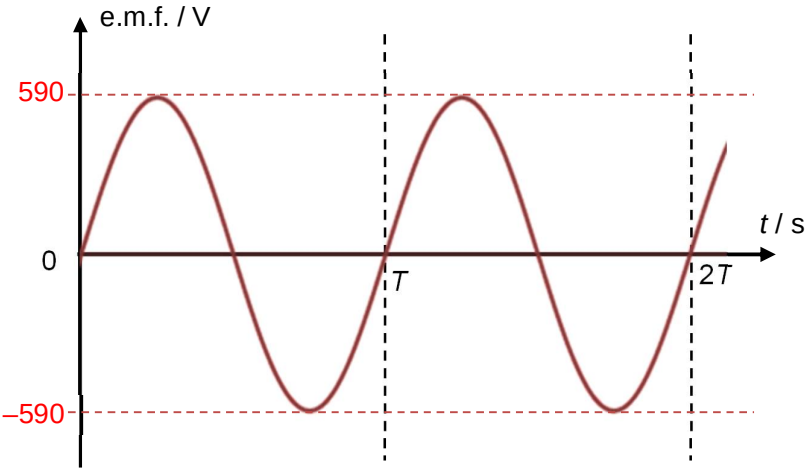
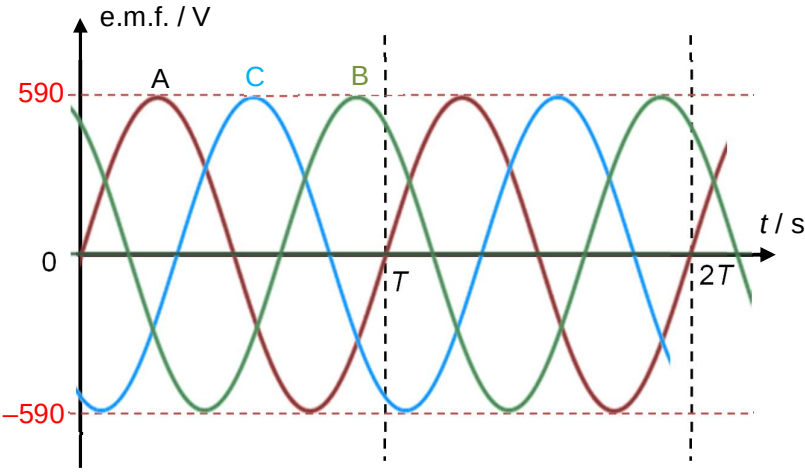


8	(a)		the <u>e.m.f. induced in a conductor is directly proportional to the rate of change of magnetic flux linkage</u> (or the rate of cutting of magnetic flux)	
	(b)	(i)	As the magnet rotates, the <u>magnetic flux density of the magnet changes orientation</u> (or alignment with coil or change in direction of magnetic field) which causes a change in magnetic flux linkage with the coil. With a <u>constant rotational rate/speed/frequency</u>, the <u>magnetic flux linkage with the coil will vary sinusoidally with time</u> (with $\cos\omega t$), By Faraday's law, the induced e.m.f. in the coil, which is the <u>rate of change of magnetic flux linkage</u>, will also <u>vary sinusoidally with time</u> (with $-\sin\omega t$).	
		(ii)	$\Phi = NBA = 500(0.80)(\pi(0.05)^2) = \pi = 3.14159 \text{ Wb (or Wb turns)}$ Correct substitution – C1 Correct answer – A1 Correct units – B1	
		(iii)	$\Phi = NBA\cos\left(\frac{2\pi}{T}t\right)$ $= 3.14\cos\left(\frac{2\pi}{\frac{60}{1800}}t\right)$ $= 3.1 \cos(60\pi t) \text{ or } 3.1 \cos(190 t)$	
		(iv)	$E = -\frac{d\Phi}{dt} = NBA\omega\sin\omega t$ $\text{max} = NBA\omega = \pi\left(2\pi\left(\frac{1800}{60}\right)\right)$ $= 592 \text{ V}$	
		(v)	 Correct +ve sine curve with at least two cycles drawn Labelled maximum induced emf of value in (iv) of 590 V (allow ecf) Maximum induced emf is <u>where the rate of change of magnetic flux linkage is greatest</u>, which is where gradient is steepest on the graph in Fig. 8.4	

(c)		 Two additional sinusoids that are roughly 1/3 of a cycle displaced from each other, and are of equal frequency and amplitude as each other Correct labelling of A, B and C (allow labelling as AA', BB', CC') (Allow e.c.f. if original A graph is wrongly drawn as -ve sine)	
(d)	(i)	Power transfer from primary coil to secondary coil is 100% efficient (or no loss in power when transferring from primary to secondary) Or no hysteresis, etc.	
	(ii)	Since transformer is ideal, $\frac{V_s}{V_p} = \frac{N_s}{N_p}$ $V_s = \frac{12000}{30} (590) = 236000 \text{ V (peak)}$ peak power at primary coil = (20)(590) = 11800 W peak current at secondary coil = 11800 / 236000 = 0.050 A r.m.s current in secondary coil = 0.050 / $\sqrt{2}$ = 0.03545 = 0.035 A OR $\frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s}$ $I_s = \frac{30}{12000} (20) = 0.050 \text{ A}$ r.m.s current in secondary coil = 0.050 / $\sqrt{2}$ = 0.03545 = 0.035 A	
	(iii)	Long distance transmission the resistance of the cables become significant A <u>low current</u> allows for <u>low power dissipation</u> (loss) in the transmission cables	

END OF PAPER